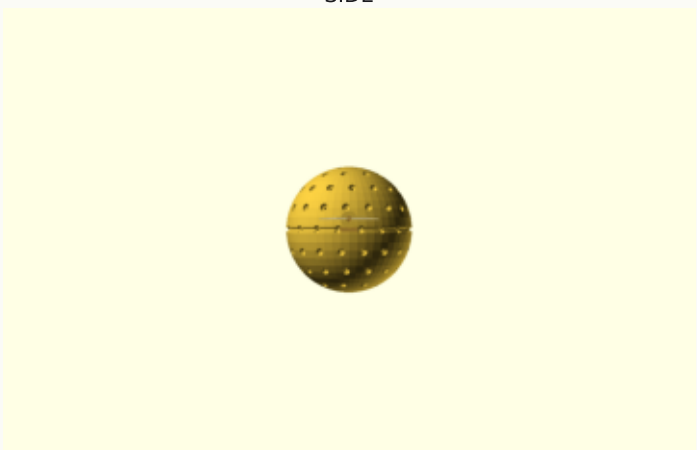


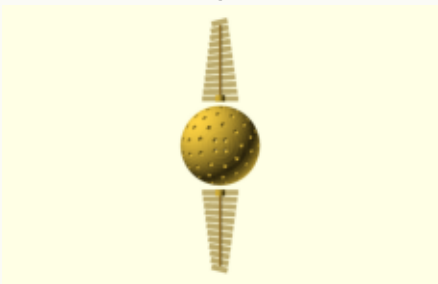
ISOMETRIC



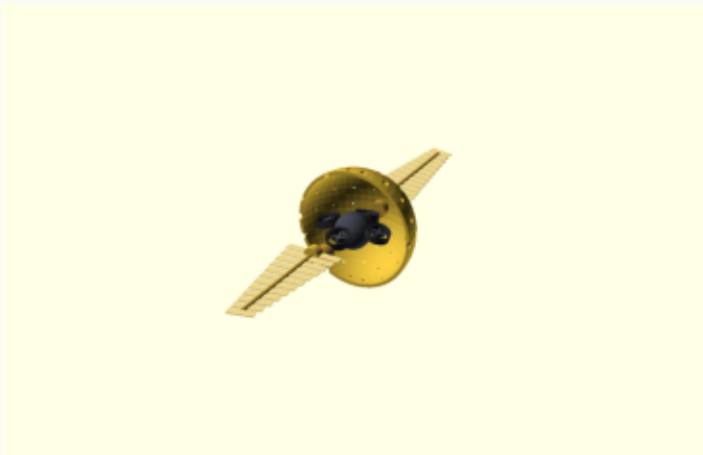
SIDE



TOP



CUTAWAY — internal drive revealed



KEY DIMENSIONS & SPEC

(every number ties back to the flight software)

Shell radius40 mm (Ø80)

ball/params.py snitch.radius = 0.04 m

Mass target50 g

snitch.mass = 0.05 kg

Top speed19 m/s

snitch.max_speed

Max accel32 m/s²

snitch.max_accel (absurdly agile)

Lift4 × micro-EDF (internal)

swarm core – the real flight

Wings2 × flapping, 8-14 Hz

coreless gearmotor driven

Capturehand held 0.25 s

capture_radius 0.22 / capture_dwll

Danger sense4.0 m

snitch.danger_radius

NOTE

OFFICIAL STORY: it flies by flapping its wings. The wings are thin and genuinely flutter (SNT-02/03/04). The lift is actually the internal micro-EDF swarm (SNT-05).

Do not explain this to Potterheads.

BILL OF MATERIALS — SNT (14 line items, prototype qty 1)

REF	PART	SPEC	QTY	\$ EA	\$ EXT
SNT-01	Gold-clad vented sphere shell	Ø80 mm (R=40 mm), walnut-sized, dimple vents	1	90	90
SNT-02	Animatronic flapping wing ass...	thin filigree vane on hinge, span ~76 mm	2	28	56
SNT-03	Coreless gearmotor + crank	6 mm coreless, flap linkage, 8-14 Hz beat	2	9	18
SNT-04	Wing membrane	5 micron gold-tint mylar, feather-cut	2	4	8
SNT-05	Micro ducted-fan lift unit	Ø16 mm internal, the ACTUAL lift	4	22	88
SNT-06	Micro brushless motor	4 mm, paired to micro-EDF	4	11	44
SNT-07	Micro ESC (4-in-1)	5 A, integrated	1	26	26
SNT-08	Nano flight controller	STM32 class, IMU on-board, predictive evasion	1	38	38
SNT-09	Optical-flow + ToF sensor	360deg-ish reach detection	2	16	32
SNT-10	Micro LiDAR / proximity ring	detects incoming hands/airframes	1	45	45
SNT-11	Capacitive capture sensor	registers an enclosing hand	1	12	12
SNT-12	Micro Li-Po cell	1S 300 mAh, lightweight	1	8	8
SNT-13	Gold micro-LED	glow so the crowd can track it	1	5	5
SNT-14	2.4 GHz telemetry link	reports pose to the Dementor referee	1	14	14

UNIT TOTAL (prototype):\$484